

SmartGrid Sentinel: Predictive Load Shedding Risk Forecasting & Intelligent Alert System

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1 Project Overview

SmartGrid Sentinel is a hybrid AI-driven energy risk monitoring platform designed to forecast potential load shedding risks at the Upazila level across Bangladesh.

The system combines:

- Deep Learning-based time-series forecasting
- NLP-based alert explanation
- Event-driven warning mechanisms

to provide localized predictive electricity risk awareness for citizens, businesses, and administrators.

The platform uses:

- real Bangladesh administrative data,
- real historical weather data,
- and simulated electricity demand behavior

to model load shedding risk patterns in a scalable and intelligent way.

2 Problem Statement

Bangladesh frequently experiences electricity instability during periods of high demand, extreme weather conditions, and grid stress. However, publicly accessible localized forecasting and intelligent warning systems are limited.

Most existing systems:

- provide reactive information,
- lack predictive intelligence,
- do not operate at localized administrative levels,
- and fail to explain risks clearly to end users.

As a result, people and businesses cannot prepare proactively for potential power disruptions.

3 Proposed Solution

The proposed system introduces an Upazila-level predictive monitoring platform capable of:

- forecasting future load shedding risk,
- generating intelligent warning alerts,
- and translating AI outputs into human-readable explanations.

The platform combines:

- LSTM-based forecasting,
- NLP-driven explanation generation,
- and event-triggered alert mechanisms

within a modern web-based dashboard architecture.

4 Objectives

4.1 Primary Objectives

- Predict load shedding risk using deep learning techniques
- Provide Upazila-level risk forecasting
- Generate intelligent warning alerts
- Build a scalable smart-grid inspired monitoring platform

4.2 Secondary Objectives

- Improve interpretability of AI predictions
- Support proactive decision-making
- Demonstrate hybrid AI architecture
- Simulate realistic smart energy monitoring workflows

5 Scope of the Project

The system focuses on:

- Bangladesh administrative regions
- Upazila-level forecasting
- Time-series energy risk prediction
- Weather-aware demand modeling
- AI-generated alert explanation

The project does not include:

- direct national grid integration,
- real-time smart meter collection,
- or actual electricity control systems.

6 Dataset Description

The dataset is a synthetic smart-grid simulation dataset built using:

- real Bangladesh administrative data,
- real weather observations,
- and simulated electricity demand behavior.

6.1 Dataset Characteristics

Property	Details
Time Range	Apr 13 – May 13, 2026
Interval	2-hour
Upazilas	477
Districts	64
Divisions	8
Total Rows	181,536
Null Values	0

6.2 Dataset Features

Feature	Description
datetime	timestamp
division	division name
district	district name
upazila	upazila name
temperature	real weather temperature
humidity	real humidity
rainfall	real rainfall
demand_index	simulated electricity demand
risk_level	target label

7 System Architecture

Administrative Data + Real Weather Data

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Demand Simulation Engine

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Dataset Preparation

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LSTM Forecasting Model

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Risk Prediction Output

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Event-Driven Alert Engine

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NLP Explanation Generation

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Web Dashboard

8 Technologies Used

8.1 AI / Deep Learning

- Python
- TensorFlow / Keras
- LSTM Networks

- Scikit-learn

8.2 NLP Layer

- Rule-Based Natural Language Generation (NLG)
- Python

8.3 Data Processing

- Pandas
- NumPy
- GeoPandas

8.4 Frontend

- Next.js
- TypeScript
- Tailwind CSS
- Framer Motion

8.5 Backend

- FastAPI / Node.js API
- REST API architecture

9 Deep Learning Model

The core forecasting engine uses:

Long Short-Term Memory (LSTM)

because:

- electricity demand is sequential,
- weather influence is temporal,
- and load patterns depend on historical behavior.

10 NLP-Based Explanation Engine

The NLP layer converts prediction outputs into:

- human-readable alerts,
- contextual warnings,
- and interpretable explanations.

Example:

High load shedding risk expected in Kaliakair Upazila due to elevated evening demand and increased temperature.

11 Event-Driven Alert System

The alert engine continuously evaluates prediction results and automatically triggers warning events when predefined thresholds are exceeded.

- High Risk → Critical Alert
- Medium Risk → Warning Alert
- Low Risk → Stable Status

12 User Types

User Type	Responsibilities
General Users	View forecasts and alerts
Business Users	Monitor operational electricity risks
Admin	Manage system monitoring and analytics

13 Expected Outcomes

- Forecast localized load shedding risks
- Improve public awareness
- Demonstrate intelligent energy monitoring
- Showcase practical hybrid AI integration

14 Innovation and Uniqueness

- Upazila-level forecasting architecture
- Hybrid AI design (LSTM + NLP + Alert Engine)
- Real-weather integrated demand simulation
- Intelligent explainable alert generation
- Bangladesh-focused smart-grid monitoring system

15 Future Enhancements

- real-time weather streaming
- mobile push notifications
- GIS visualization
- smart meter integration
- transformer-based forecasting models
- reinforcement learning optimization

16 Conclusion

SmartGrid Sentinel presents a modern AI-driven approach for predictive energy risk monitoring in Bangladesh. By combining deep learning forecasting, intelligent alerting, and NLP-based explanation generation, the system demonstrates how hybrid AI technologies can support localized smart-grid awareness and proactive energy risk management.

17 References

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